DIPLOMA PROGRAMME PRACTICE

Mathematics

Applications and interpretation SL

Set A / Paper 2 / Extended response

90 minutes · 80 marks · All questions compulsory

Use a graphic display calculator and a clean Mathematics formula booklet.

Write your reasoning and working in a separate answer booklet. Unless stated otherwise, give exact answers or round numerical results to three significant figures.

Diagrams are illustrative unless coordinates or dimensions are specified.

QUESTION 1

20 marks

A theatre needs a lighting console for three years. It can buy one today with a loan for 12 000 dollars at 4.8% nominal annual interest compounded monthly. The loan is repaid in 36 equal end-of-month instalments. Alternatively, it can rent for 290 dollars in month 1, rising by 1% each month. The console can be sold immediately after month 36 if purchased. Ignore tax, maintenance and the time value of cash when comparing totals.

(a) Find the monthly loan payment and the total of all repayments. Give monetary totals to the nearest dollar and the payment to the nearest cent. [4]

(b) Write the rent in month n and find the total rent for the 36 months. [4]

(c) Determine the first month in which that month's rent exceeds one loan instalment. Justify the discrete answer. [4]

(d) Determine the break-even resale price after 36 months. A dealer offers a guaranteed 1050 dollars. Recommend one option using the numerical margin, not just the break-even figure. [5]

(e) Rental includes free replacement during a breakdown; a purchased console has no such cover. Explain how to use the result above and an estimate of the cost of lost performances to reassess the decision. [3]

QUESTION 2

20 marks

At a harbour the modelled water depth during a 12-hour cycle is $d(t) = 3.6 + 1.4\cos(30t°)$ metres, where t is hours after 06:00 and $0 \le t \le 12$. A ferry requires at least 3.0 m to enter. Treat the model as continuous and use degrees.

(a) Find the maximum depth, the minimum depth and the period. Explain the role of 3.6 in the model. **[4]**

(b) Determine the two times when the depth is exactly 3.0 m. Give clock times to the nearest minute. **[4]**

(c) The ferry must stay above or at the threshold for its entire 45-minute approach. Determine the latest starting time, to the nearest minute, for an approach completed before the first unsafe interval. Explain why testing depth only at departure is inadequate. **[4]**

(d) At $t = 0$, 2, 4 and 6 h, sensors record outgoing current speeds of 0.2, 0.8, 1.1 and 0.5 km/h. Estimate the distance a freely drifting marker would travel in these six hours, using the trapezoidal rule. State an assumption. **[4]**

(e) A cross-check shows every depth prediction may overstate actual depth by up to 0.20 m. Departures are scheduled only at whole minutes. Find the latest start for the same first approach that guarantees safe clearance. **[4]**

QUESTION 3

20 marks

A transport authority takes an independent simple random sample of 160 commuters from its register. Among the 80 sampled from the north district, 46 used the bus, 24 cycled and 10 walked. Among the 80 from the south district, the counts were 30, 32 and 18. Each commuter reports exactly one main mode. Use a 5% significance level.

(a) State the null and alternative hypotheses for a chi-squared test of independence and calculate the column totals. **[3]**

(b) Calculate all expected frequencies under H0. Explain why the usual expected-frequency condition is met. **[4]**

(c) Use technology to calculate the test statistic and p-value. State the degrees of freedom and draw a conclusion. **[5]**

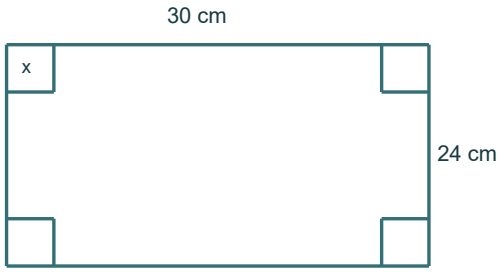
(d) For a randomly chosen sampled commuter who used the bus, find the probability that they live in the north. Compare the observed fraction of bus users in the north district with that in the south district. **[4]**

(e) A planner claims that moving a person north will make them use the bus and that the result describes every traveller in the city. Evaluate both claims. **[4]**

QUESTION 4

20 marks

A rectangular sheet 30 cm by 24 cm is made into an open tray. Equal squares of side x cm are cut from its four corners. The flaps are folded vertically without overlap. Neglect thickness.



Corner cuts (not to scale)

(a) Express the two base dimensions in terms of x and state the physical domain for a tray with positive volume. **[3]**

(b) Show that $V(x) = 4x^3 - 108x^2 + 720x$ cm³. Find its derivative and both stationary x-values. **[5]**

(c) Determine the maximum physical volume and justify why your stationary point is the maximum. **[4]**

(d) The cutter can use only increments of 0.5 cm. Choose the best allowable cut and calculate the lost capacity relative to the continuous optimum. **[4]**

(e) A liner reduces the usable volume of each 4.5 cm tray by 6%. A caterer orders 12 litres of total usable capacity. Determine the minimum number of trays and justify the rounding direction. **[4]**